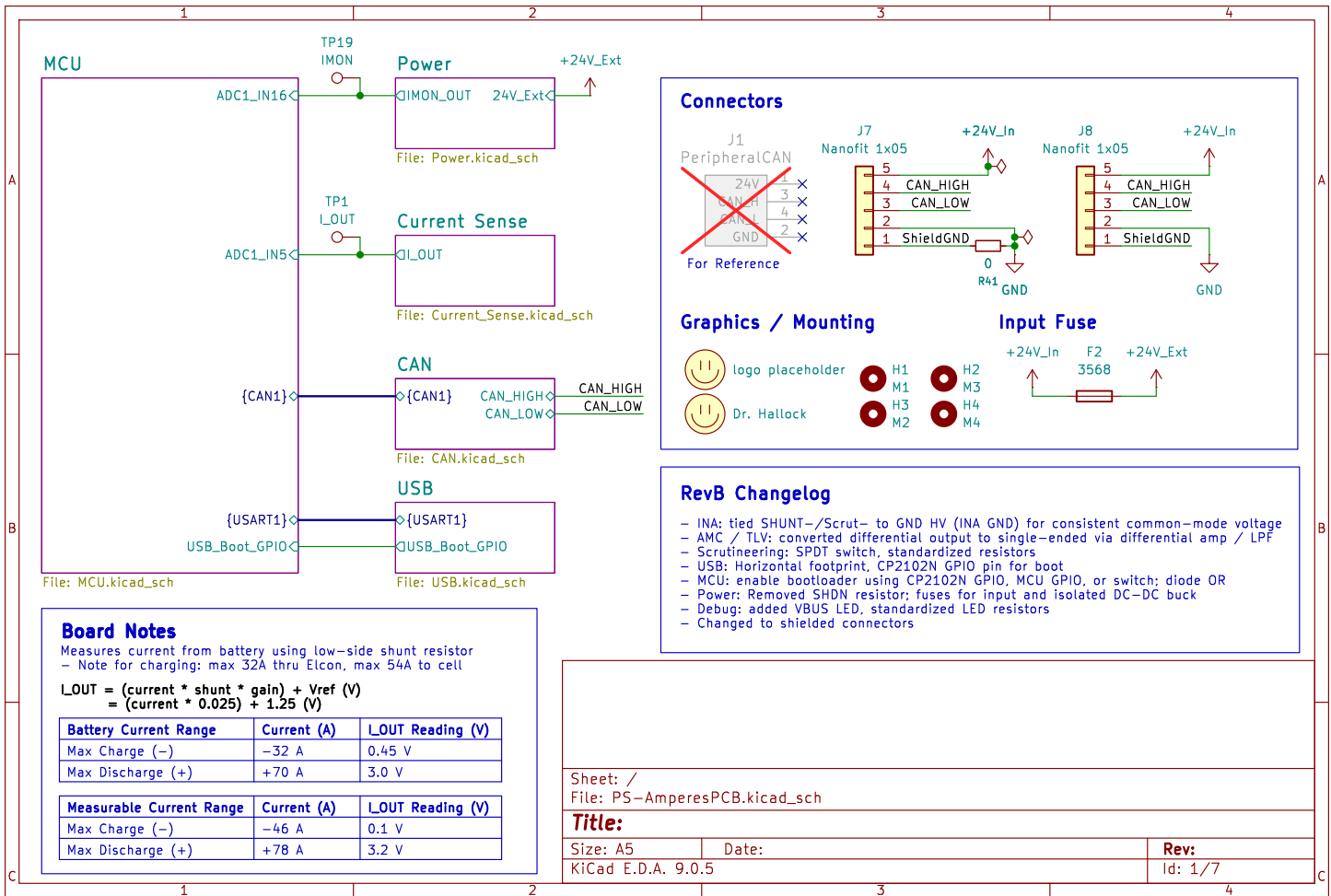


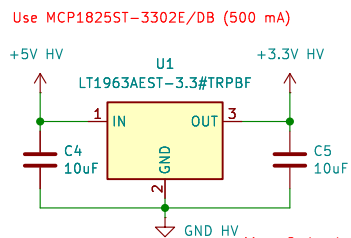
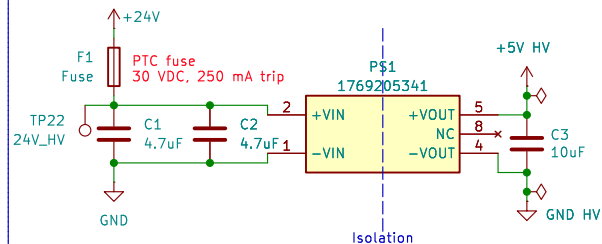


Power

3.3V HV (Isolated)

24V → 5V Buck

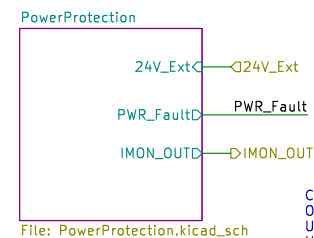
5V → 3.3V LDO



Max Output Current: 0.2A

Power Protection

Taken from PSOM

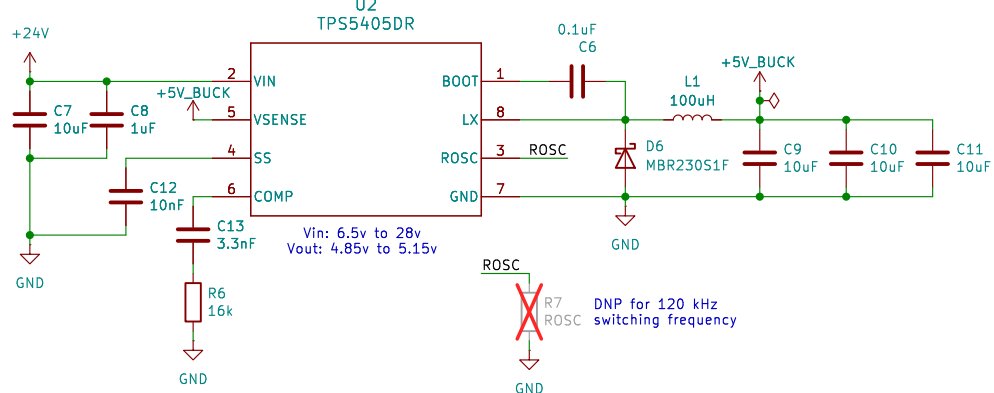


Current Limit: 1.2A
Overvoltage: ~27v
Undervoltage: ~20v
Use 1% resistors
for closest values

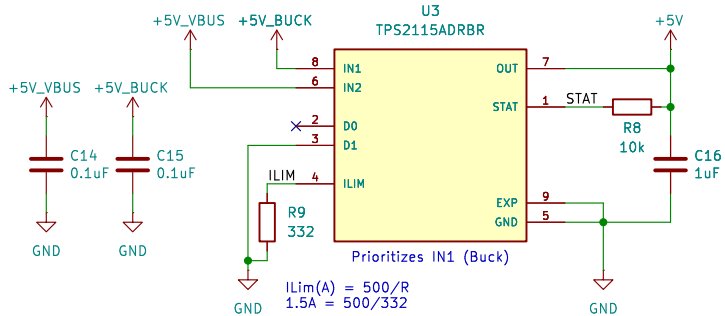
3.3V LV (MCU, CAN, etc)

Taken from PSOM

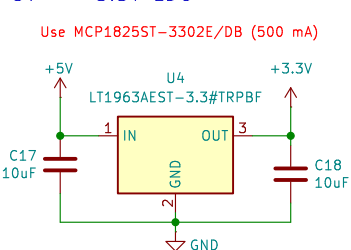
24V → 5V Buck



5v Power Mux



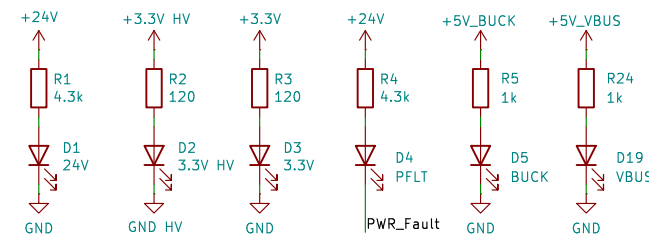
5V → 3.3V LDO



Max Output Current: 1.5A

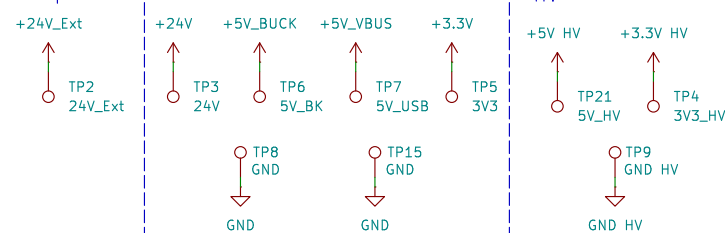
Indicators

Fault active low



Testpoints

LV



Sheet: /Power/
File: Power.kicad_sch

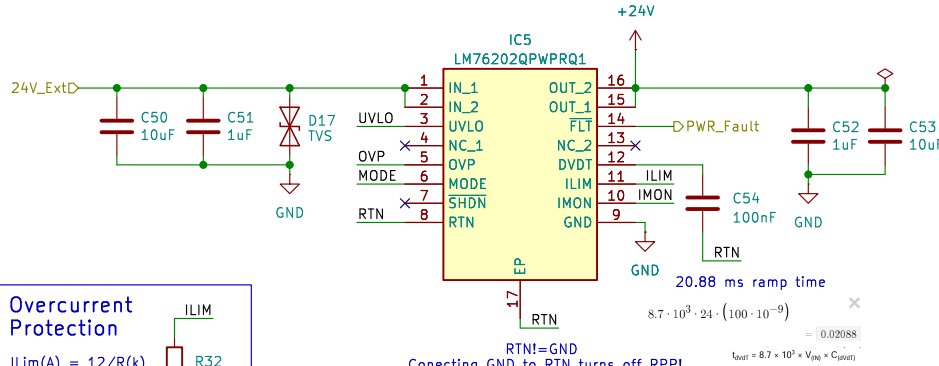
Title:

Size: A4
KiCad E.D.A. 9.0.5

Date:

Rev:
Id: 2/7

Taken from PSOM



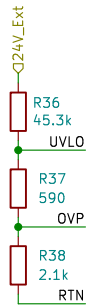
Overcurrent Protection

$$I_{Lim}(A) = 12/R(k)$$

$$1.2A = 12/10k$$



Under/Overvoltage Protection



	Setpoint	Ideal	Actual
UVLO	20v	1.1v	1.12v
OVP	27v	1.1v	1.18v

OVP set to 27v for buck max voltage of 28v

$$\left(\frac{2.1 + 0.59}{2.1 + 0.59 + 45.3} \right) 20$$

$$= 1.1210668894$$

$$\left(\frac{2.1}{2.1 + 0.59 + 45.3} \right) 27$$

$$= 1.18149614503$$

Functional Mode Selection

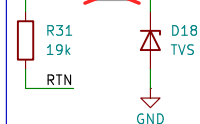
Set to circuit breaker (latchoff)

Use 0 ohm resistor for auto-retry (540ms)



Current Monitoring

R30 DNP until validated



Typical gain: 78.28 uA/A
Vout @ 1.3A = 2.407v
R(imon) = 23.7 mΩ

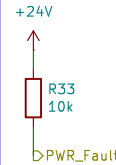
$$R_{(IMONmax)} = \frac{\min[(V_{in} - 1.5), 4 V]}{1.8 \times I_{(LIM)} \times GAIN(IMON)}$$

$$\text{For } I_{(LIM)} > 50 \text{ mA, } V_{(IMON)} = [I_{(OUT)} \times GAIN(IMON)] \times R_{(IMON)}$$

$$\text{For } I_{(LIM)} < 50 \text{ mA (typical), IMON output current is close to } I_{(MAX,OUT)}$$

$$\text{With } R_{(IMON)} = (R_{(MAX, OUT)} - R_{(IMON)})$$

Pullups



RIP shdn resistor

Sheet: /Power/PowerProtection/
File: PowerProtection.kicad_sch

Title:

Size: A5

Date:

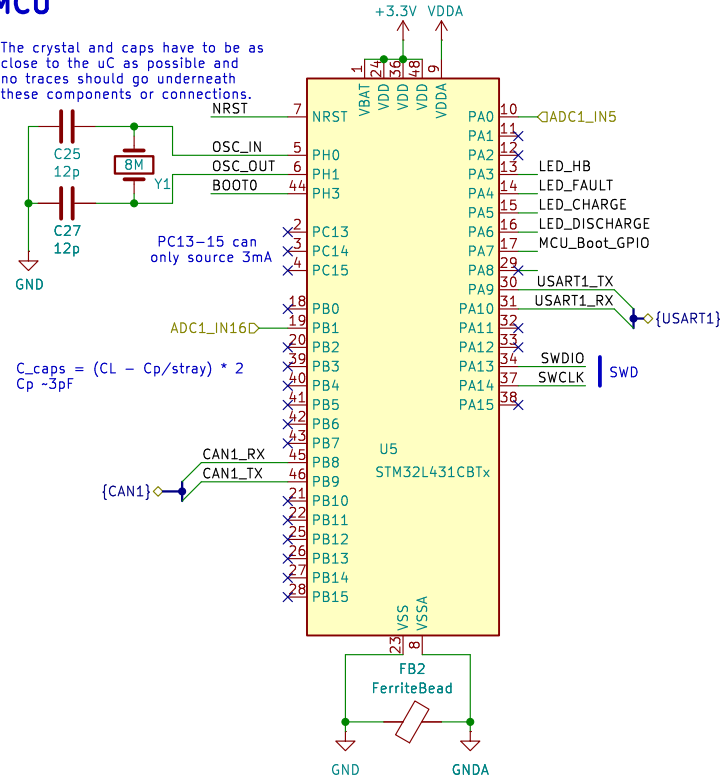
KiCad E.D.A. 9.0.5

Rev:

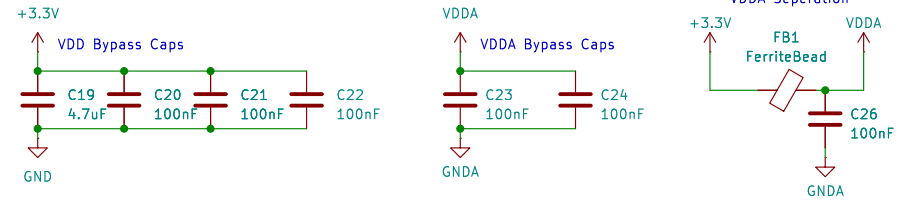
Id: 7/7

MCU

The crystal and caps have to be as close to the uC as possible and no traces should go underneath these components or connections.

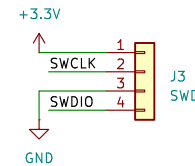


Bypass, Separation

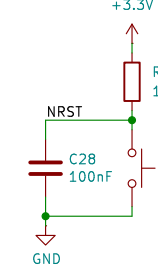


SWD Connector

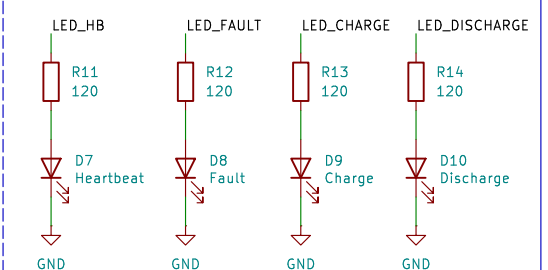
Moxie Nanofit 1053091104
Mates with 0022013047



Reset Button



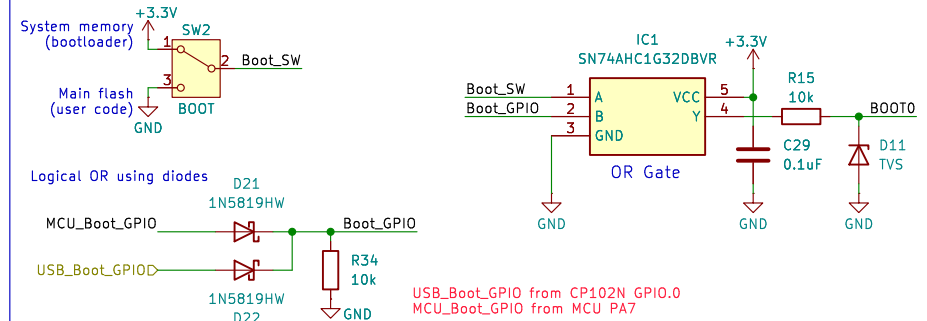
LED Indicators



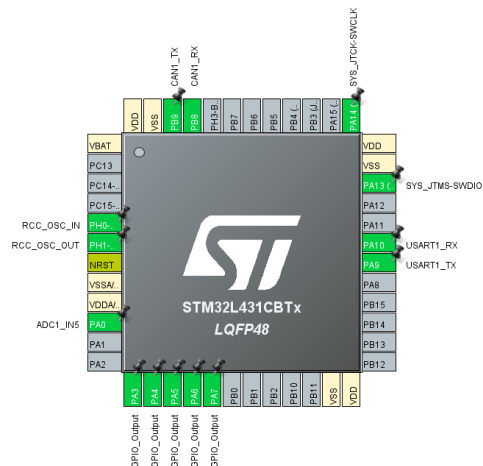
USART bootloader pins (same as PSOM)

USART1_RX pin	Input	PA10 pin: USART1 in reception mode. Used in input no pull mode.
USART1_TX pin	Output	PA9 pin: USART1 in transmission mode. Used in input no pull mode.

Boot Selector



Pin Functions



Sheet: /MCU/
File: MCU.kicad_sch

Title:

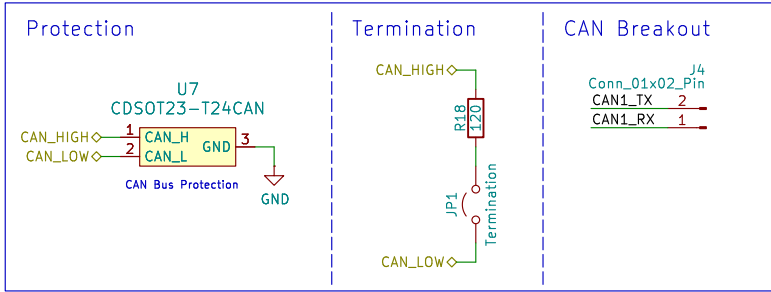
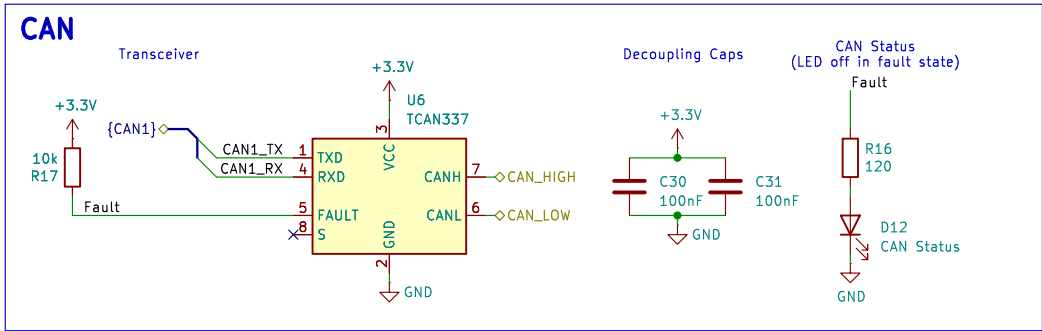
Size: A4

Date:

KiCad E.D.A. 9.0.5

Rev:

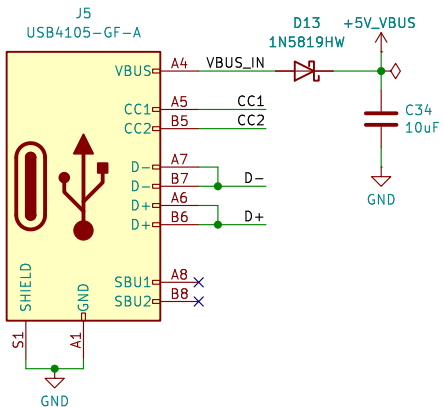
Id: 3/7



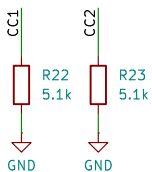
Sheet: /CAN/ File: CAN.kicad_sch		
Title:		
Size: A5	Date:	Rev:
KiCad E.D.A. 9.0.5		Id: 4/7

USB-C

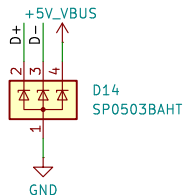
Horizontal USB C Port



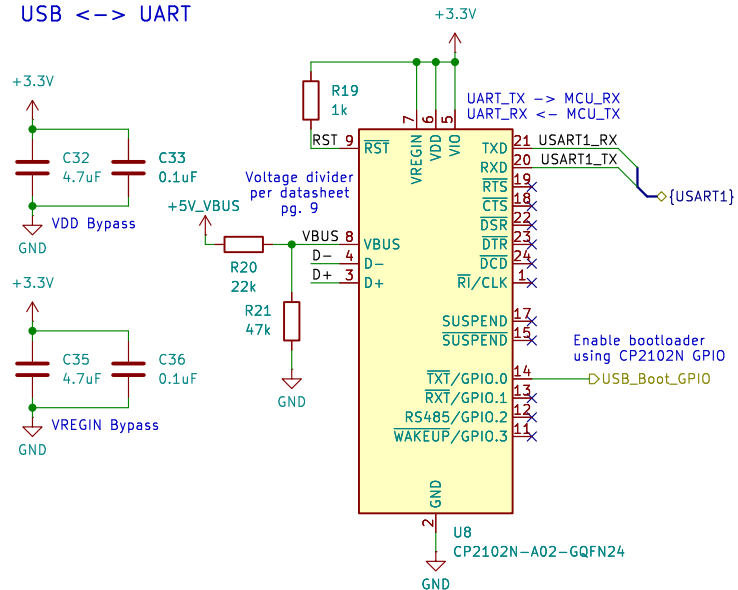
CCx Pulldown



USB Protection



USB <-> UART



Sheet: /USB/
File: USB.kicad_sch

Title:

Size: A5
KiCad E.D.A. 9.0.5

Date:

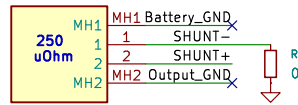
Rev:

Id: 5/7

Input

250uOhm Shunt Resistor

R39
WSBS8518L2500JTP3

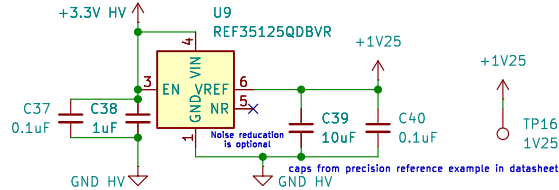


Footprint is M4 version; edit it to have M3 holes

SHUNT- SHUNT- TP11
SHUNT+ SHUNT+ TP10

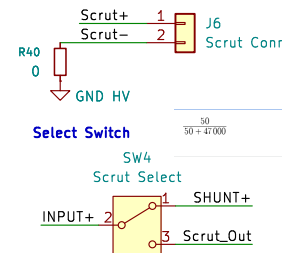
Expected current range: -32 to 70 A
w/ corr. voltage range: -8 to 17.5 mV
Low-side shunt, so current path is as follows:
Discharge: Output_GND -> Battery_GND
Charge: Battery_GND -> Output_GND

Voltage Reference (1.25V)

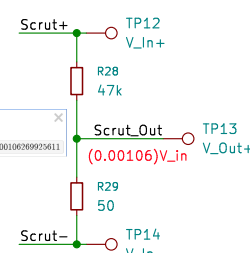


Current Scrutineering

Input



Divider



DIRECTIONS

- Pass in voltage from scrutineering board
- Use switch to select between shunt / scrutineering inputs

RELATIONSHIP

- * $L_{OUT} = ((0.00106)V_{in} * 100) + 1.25 (V)$
- * Spoof Current = $(0.00106)V_{in} / (250 \text{ uOhm}) (A)$

Example

Input (V)	Scrut_Out (mV)	Spoof Current (A)	L_OUT Reading (V)
-10 V	-10.6 mV	-42.5 A	0.19 V
+17 V	+18 mV	+72.2 A	3.05 V

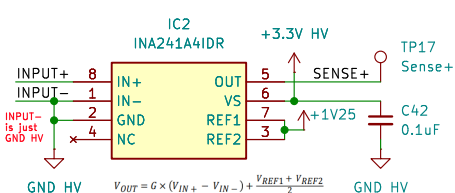
1% tolerance on resistors in voltage divider

SHUNT- and Scrut- connected to GND HV to prevent common-mode voltage from being out of INA's range

Current Sense Path

Shunt -> current sense amp -> isolation -> filter

Bidirectional current sense amp



Gain: 100 V/V (INA241 A4 version)

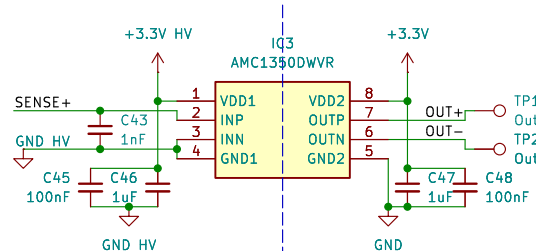
Reference voltage (bias for discharge): 1.25V

1. $(70A * 250 \text{ uOhm} * 100V) + V_{ref} = 3 \text{ V}$
2. $(-32A * 250 \text{ uOhm} * 100V) + V_{ref} = 0.45 \text{ V}$

Maximum current range: -46 to +78A (0.1 to 3.2V output)

Make sure common-mode stuff is within range

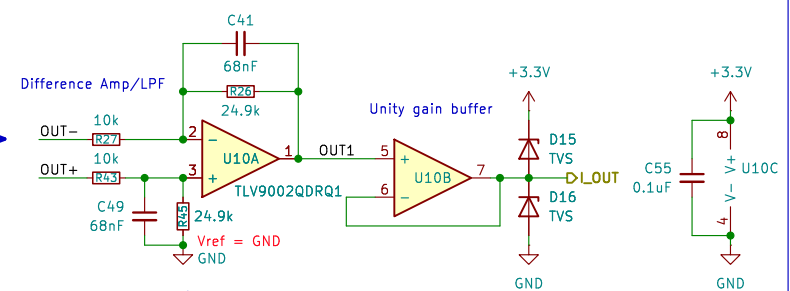
Isolation



Fixed Gain: 0.4 V/V
Target Voltage Ranges:
* Input: 0.1 - 3.2 V
* Output: 0.04 - 1.28 V

OUTN should NOT be tied to GND

Amplification + Low-Pass Filter



Differential Amplifier / Active LPF

- * $V_{out} = [(OUT+) - (OUT-)] * Gain$
- * Gain = 2.49 V/V
- * Cutoff freq = ~94 Hz
- * 1st order lowpass: -20 db/dec rolloff

See LPF documentation in Amperes Design folder

Target Voltage Ranges:
* Input: 0.04 - 1.28 V
* Output: 0.01 - 3.19 V

Table 8-1. R_SENSE Selection and Power Dissipation⁽¹⁾

PARAMETER	EQUATION	RESULTS AT V _{IN} = 5V				
		A1, B1 DEVICES	A2, B2 DEVICES	A3, B3 DEVICES	A4, B4 DEVICES	A5, B5 DEVICES
G	Gain	100V/V	200V/V	500V/V	1000V/V	2000V/V
V _{SENSE}	Ideal differential input voltage	$V_{SENSE} = V_{OUT} / G$	500mV	250mV	100mV	25mV
R _{SENSE}	Current sense resistor value	$R_{SENSE} = V_{SENSE} / I_{MAX}$	50mΩ	25mΩ	10mΩ	2.5mΩ
P _{SENSE}	Current-sense resistor power dissipation	$P_{SENSE} = I_{MAX}^2 R_{SENSE}$	0W	2.5W	1W	0.5W

Gain: 100 V/V
R_{sense} = 250 uOhm
P_{sense} = R_{sense} * (70A)² = 1.225 W

Picked shunt first, then adjusted gain to meet ADC range
If additional gain is needed, tweak filter circuit

Current to L_OUT Conversion

$$L_{OUT} = (\text{current} * \text{shunt} * \text{gain}) + V_{ref} (V)$$

ADC to Current Conversion

$$\text{Current} = ((ADC * (3.3 / 4096)) - V_{ref}) / (\text{Shunt} * \text{gain}) (A)$$

V_{ref} = 1.25 V | Shunt = 250 uOhm | Gain = -100 V/V (100 * 0.4 * 2.49)
Shunt * gain = 250 uOhm * 100 V/V = 0.025

Sheet: /Current Sense/

File: Current_Sense.kicad_sch

Title:

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